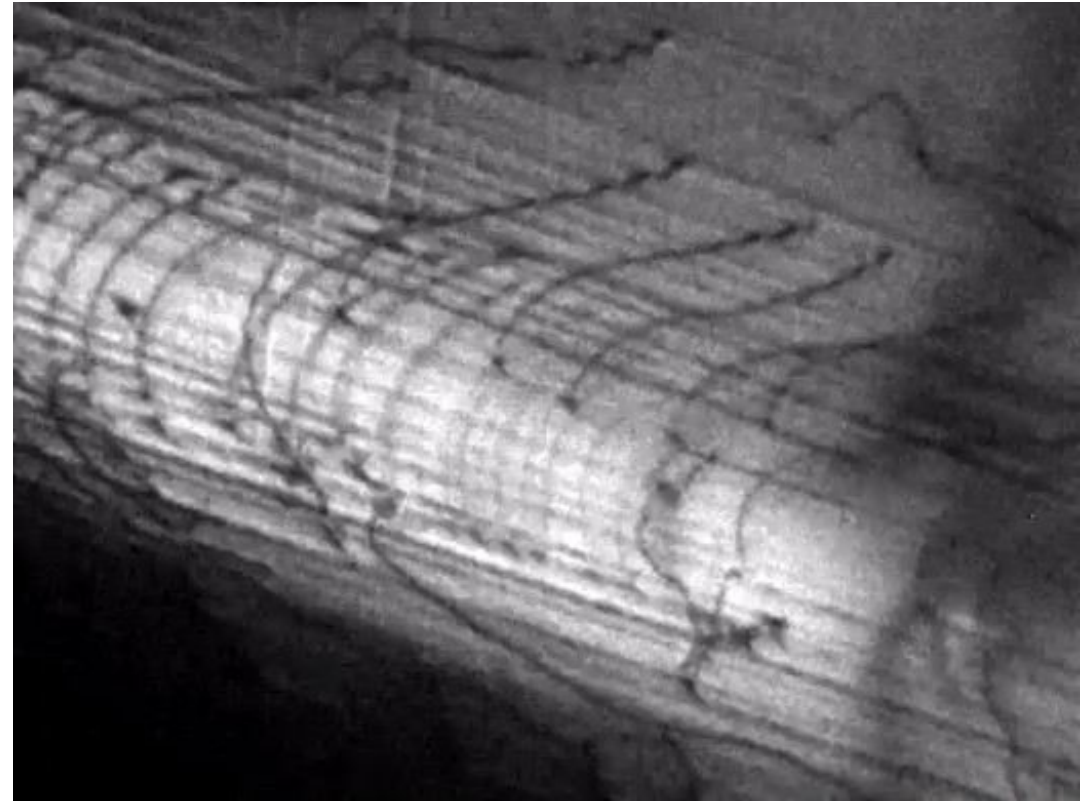
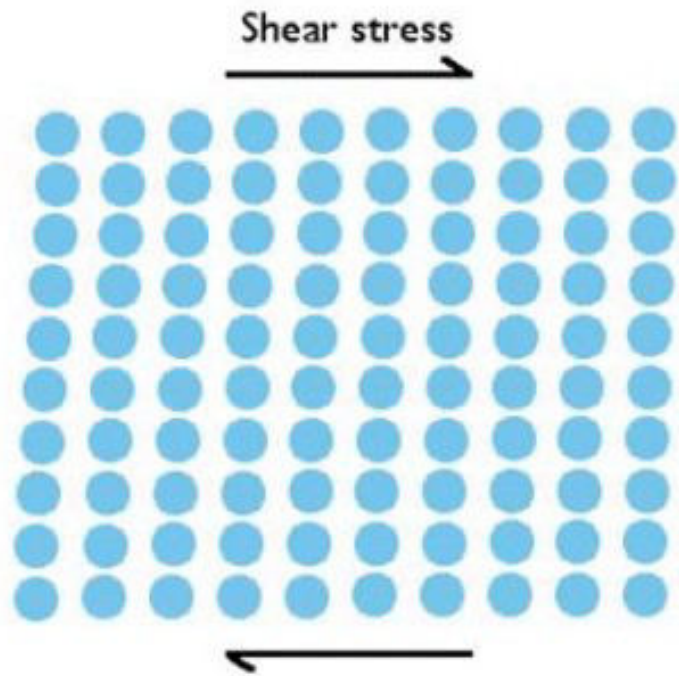


APS110 Tutorial #7

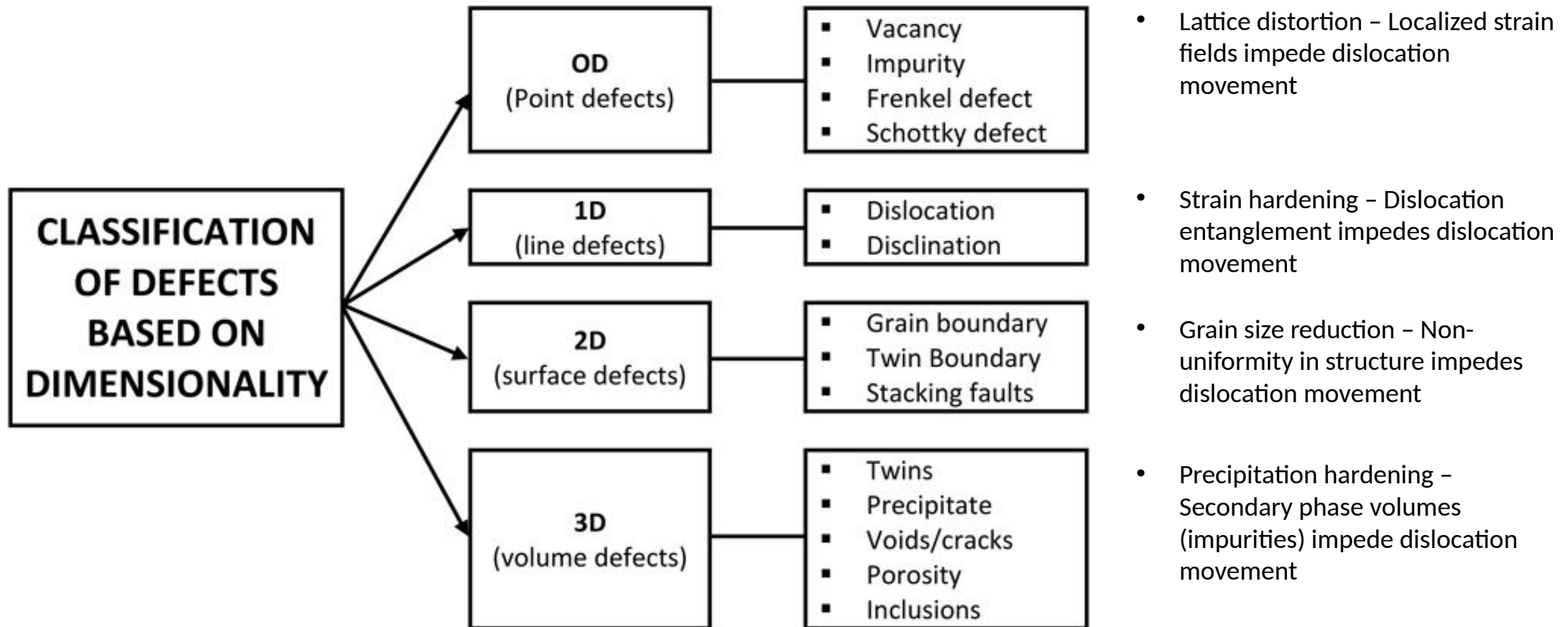
Daniel Brlas

Email: daniel.brlas@mail.utoronto.ca

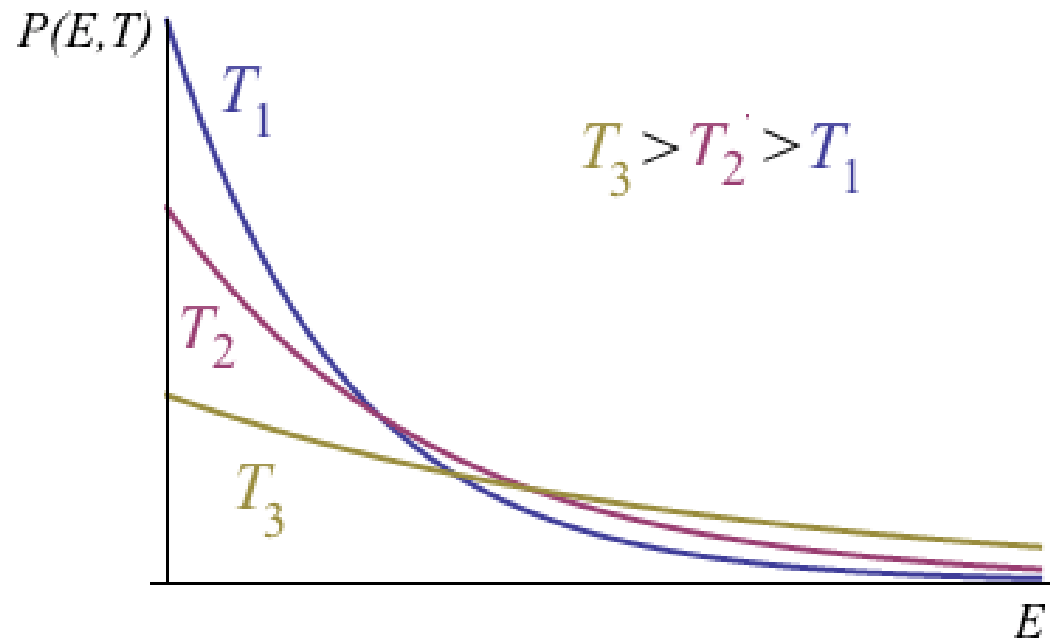
Review – How Materials Deform



Review – Defects



Review – Boltzmann Distribution



$$\frac{N_W}{N} = \exp\left(-\frac{Q_W}{kT}\right)$$

$$k = 1.381 \times 10^{-23} \frac{J}{K} \text{ or } 8.617 \times 10^{-5} \frac{eV}{K}$$

Practice Question 1

1.

- a) Calculate the fraction of atom sites that are vacant for copper (Cu) at its melting temperature of 1084°C. Assume an energy for vacancy formation of 0.90 eV/atom.
- b) Repeat this calculation at room temperature (25°C). What is the ratio of N_v/N at the melting temperature vs. room temperature?

$$\begin{aligned} \text{a) } \frac{N_v}{N} &= \exp\left(\frac{-Q_v}{kT}\right) \\ &= \exp\left[\frac{-0.90 \text{ eV/atom}}{8.62 \times 10^{-5} \cdot 1357}\right] \\ &= 4.56 \times 10^{-4} \end{aligned}$$

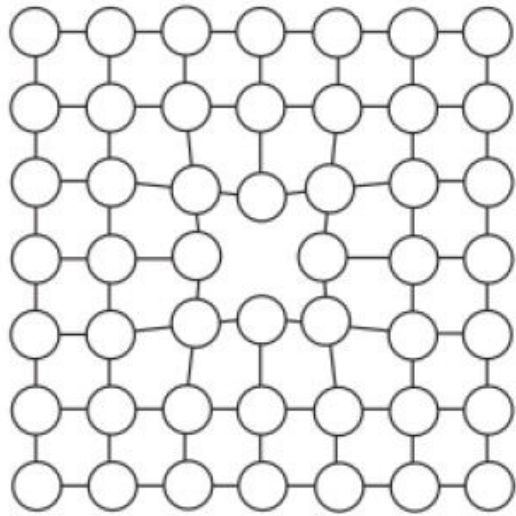
$$\begin{aligned} \text{b) } \frac{N_v}{N} &= \exp\left(\frac{-Q_v}{kT}\right) \\ &= \exp\left[\frac{-0.9}{8.62 \times 10^{-5} \cdot 298}\right] \\ &= 6.079 \times 10^{-16} \end{aligned}$$

$$\frac{4.56e-4}{6.079e-16} = 7.50e11$$

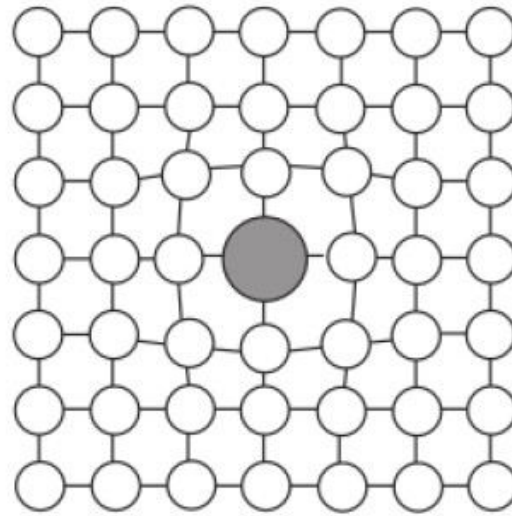
Way more vacancies at higher temperatures

Practice Question 2

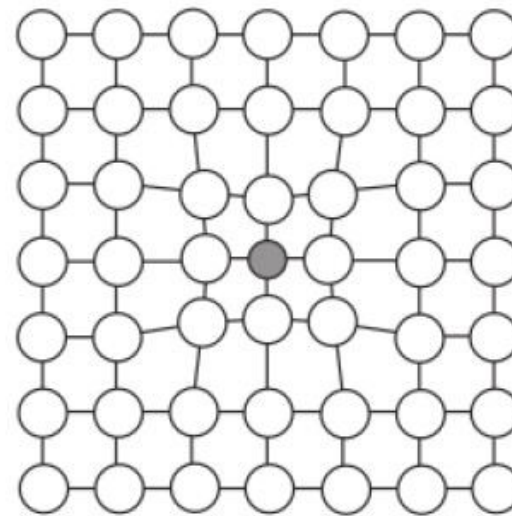
2. For each of the four images shown below, state the dimensionality and type of imperfection shown. Briefly describe what effects each defect has on the local environment.



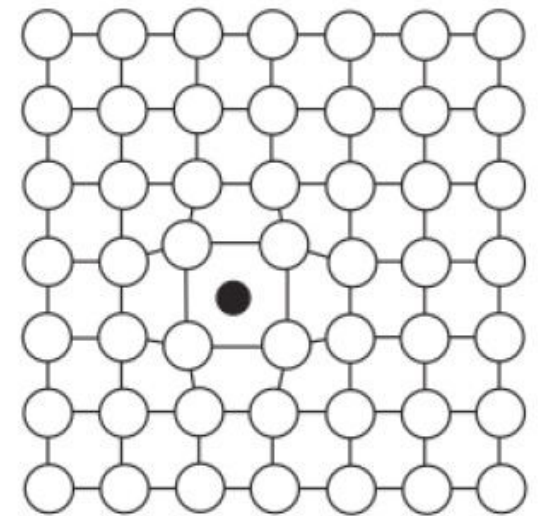
(a)



(b)



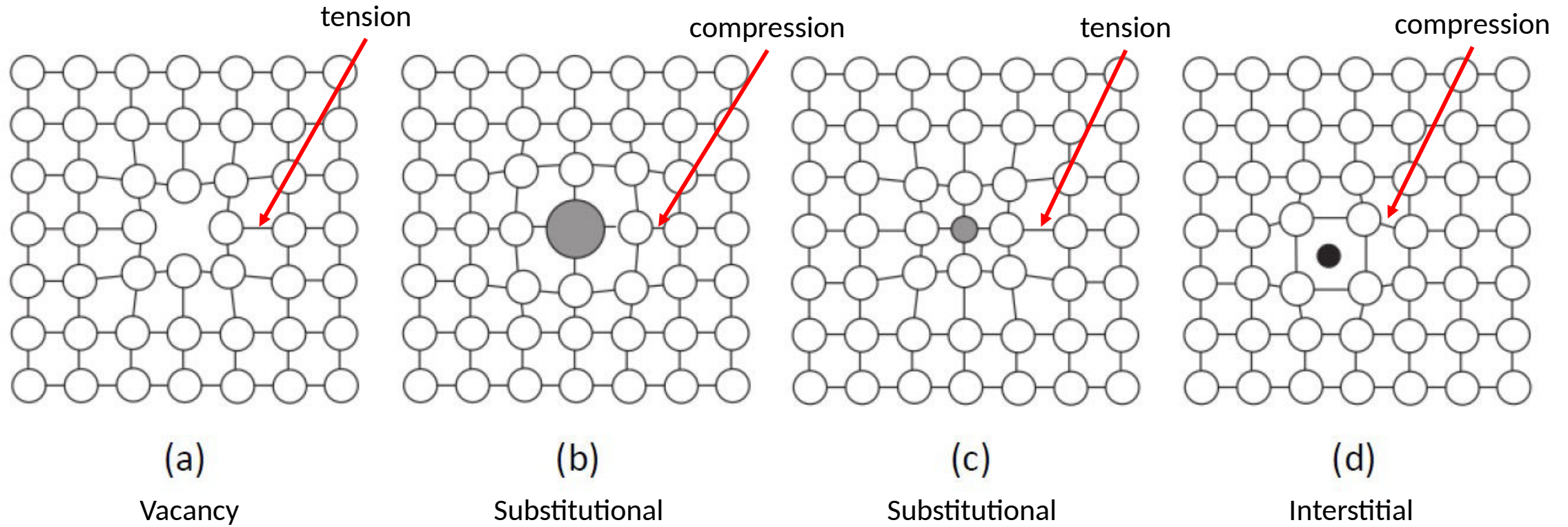
(c)



(d)

Practice Question 2

All 0D (Point) Defects



Practice Quiz

1. Copper and nickel are next to one another on the periodic table, have very similar atomic radii and both form the FCC crystal structure. What type of crystalline imperfection is a Ni atom likely to occupy within a Cu lattice?
 - a) Second phase
 - b) Substitutional
 - c) Interstitial
 - d) None of the above
2. Which of the following is the correct order of interstitial sites, by size, from smallest to largest?
 - a) Tetrahedral, octahedral, simple cubic
 - b) Octahedral, simple cubic, tetrahedral
 - c) Simple cubic, Octahedral, tetrahedral
 - d) Tetrahedral, simple cubic, octahedral
3. Which of these are examples of zero-dimensional imperfections: pores, vacancies, interstitial impurities, substitutional impurities, dislocations, grain boundaries.
 - a) Pores, dislocations, and grain boundaries
 - b) Pores, vacancies, and dislocations
 - c) Vacancies and substitutional impurities
 - d) Vacancies, interstitial impurities, and substitutional impurities

Practice Quiz

5. Which of the following best describes the Boltzmann distribution for a system at infinitely high temperature?
 - a) The particles occupy only the highest energy state
 - b) The particles occupy a distribution that peaks at the highest energy state and trails off towards the lower energy states
 - c) The particles are distributed evenly across all possible energy states
 - d) The particles occupy only the lowest energy state
6. Briefly discuss the role of grain boundaries in strengthening materials. Address the mechanism of dislocation movement, and the size of grain boundaries in this regard.
7. Jake Sparrow is a pirate who hopes to earn money by buying and selling gold. He sat in on 5 minutes of an APS110 lecture and so is aware that vacancies are thermally generated. He plans to buy only gold on cold days when the number of vacancies will be lower and then to sell the same gold on hot days when the number of vacancies will be higher.
 - a) If the energy for vacancy formation in gold is 0.67 eV, and the molar mass of gold is 196.97 g/mol, Calculate the mass (in grams) per mol of gold atoms that have formed vacancies at 25°C.
 - b) Will Jake ever make any money in this way?

6.

- ① grain boundaries exist at mismatch in crystal orientation
- ② dislocation must change direction & overcome mismatch of planes, and any lattice strain
- ③ Smaller grains mean more of these transitions must occur, meaning that material will be strengthened.

7. The right side is a picture of the question answer from a past midterm or exam

Something appears to be wrong with the derivation of the equation, however. Note that N is simply 1 for a mole. I will look into this but keep in mind it's not that important.

2. (10) Jake Sparrow is a pirate who hopes to earn money by buying and selling gold. He sat in on 5 minutes of an MSE101 lecture and so is aware that vacancies are thermally generated. He plans to only buy gold on cold days when the number of vacancies will be lower and then to sell the same gold on hot days when the number of vacancies will be higher.
- a. (7) If the energy for vacancy formation in gold is 0.67 eV, and the density of gold is 19.3 g/cm³, calculate the mass (in g) of the gold atoms that have formed vacancies at 25°C.

$$N_v = N \exp\left(\frac{-Q_v}{kT}\right)$$
$$m = \frac{N \exp\left(\frac{-Q_v}{kT}\right) \cdot A}{N_A}$$
$$m = 1.5(10^{-33}) \text{ g}$$
$$m = \frac{N_v}{N_A} \cdot A$$

$N=1$ (take 1 mol (basis))
 $A=196.97 \text{ g/mol}$
 $k=8.62(10^{-5}) \frac{\text{eV}}{\text{atom} \cdot \text{K}}$
 $T=298 \text{ K}$

or other reasonable quantity.

- b. (3) Will Jake ever make any money this way? Explain.

No. because the atoms don't disappear, they move to other locations.